

Notification System

▼ Relevant source files

Backend/STUDER/src/main/java/facu/studer/DTOs/notification/NotificationResponseDTO.java

Backend/STUDER/src/main/java/facu/studer/DTOs/notification/UserNotificationResponseDTO.java

Backend/STUDER/src/main/java/facu/studer/DTOs/user/FriendStatusResponseDTO.java

Backend/STUDER/src/main/java/facu/studer/DTOs/user/UserPublicResponseDTO.java

Backend/STUDER/src/main/java/facu/studer/DTOs/user/UserSearchPageResponseDTO.java

Backend/STUDER/src/main/java/facu/studer/entities/notifications/Notification.java

Backend/STUDER/src/main/java/facu/studer/factories/NotificationFactory.java

Backend/STUDER/src/main/java/facu/studer/mappers/NotificationMapper.java

Backend/STUDER/src/main/java/facu/studer/repositories/UserNotificationRepository.java

Backend/STUDER/src/main/java/facu/studer/services/implementation/NotificationServiceImpl.java

The Notification System in STUDER provides a centralized mechanism for alerting users about social interactions, system updates, and platform activity. It utilizes a factory-driven creation process and a structured entity model to link notifications to specific domain objects like users or discussions.

Core Data Model

The system distinguishes between a generic **Notification** (the content) and a **UserNotification** (the delivery instance for a specific user).

LinkType Enum

The **LinkType** enum defines the nature of the entity that triggered the notification, allowing the frontend to determine routing logic:

- **USER** : Related to follower actions or direct messages.

- **SYSTEM** : Platform-wide announcements.
- **MESSAGE** : Specific chat updates.
- **EVENT** : Calendar or deadline alerts.
- **DISCUSSION** : Activity in forum threads.

Entities

- **Notification**: Contains the core payload including **title**, **message**, **type** (LinkedType), and **linkedId** (the ID of the entity to navigate to).

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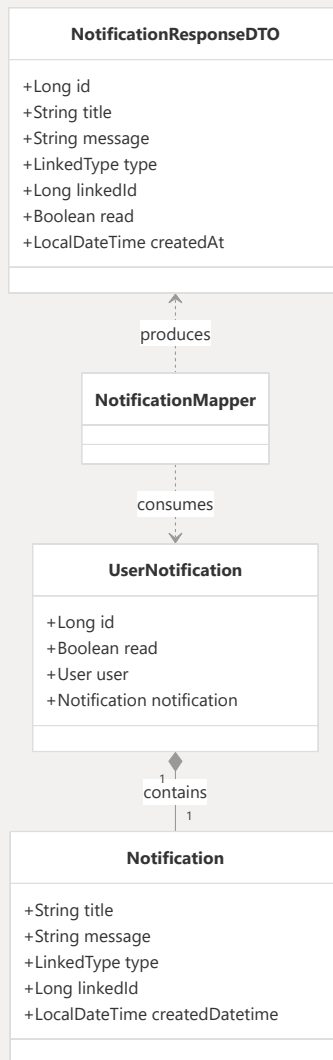
- **UserNotification**: Manages the relationship between a **User** and a **Notification**, tracking the **read** status and **readAt** timestamp.

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Notification Schema Mapping

The following diagram illustrates how the database entities map to the API response structure.

Notification Entity to DTO Mapping



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Notification Creation Logic

Notifications are instantiated using the **NotificationFactory**. This ensures consistency in id keys and property assignment across different business flows.

- **Follow Notifications:** Created when a user follows another. Sets **type** to **USER** and **linkedId** to the follower's ID.

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- **Message Notifications:** Created for new direct messages. It dynamically selects between `message.new_message` or `message.new_request` keys based on the friendship status.
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Service Implementation

The `NotificationServiceImpl` handles the retrieval and state management of notifications.

Paginated Retrieval

The `getNotifications` method uses a `Specification` to build dynamic queries based on:

1. **Username:** Filters for the authenticated user.

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2. **Type:** Optional filter for `LinkType`.

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3. **Read Status:** Filter for read/unread notifications.

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4. **Temporal Filter:** Filter by `lastDays`.

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The results are returned as a `NotificationPageResponseDTO` with a default page size of 10.

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Marking as Read

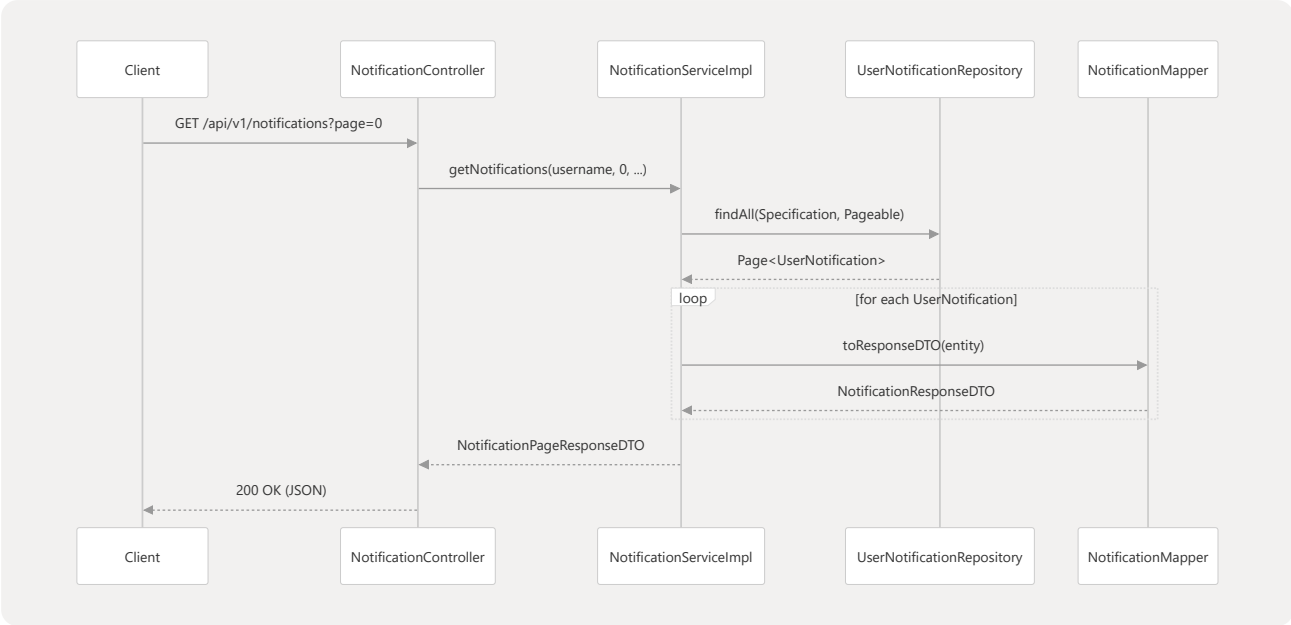
When a user interacts with a notification, the `markAsRead` method updates the `UserNotification` entity. It performs an ownership check to ensure users can only modify their own notifications.

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API Data Flow

The following diagram traces the flow from a database query to the REST response via the `NotificationController` logic.

Notification Retrieval Flow



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Key Classes and Components

Component	Responsibility
NotificationFactory	Static methods to build typed <code>Notification</code> objects with localized keys.
NotificationMapper	Converts <code>UserNotification</code> entities into <code>NotificationResponseDTO</code> .
NotificationServiceImpl	Core business logic, pagination, and JPA Specification filtering.
UserNotificationRepository	Data access layer with custom query <code>findByIdWithNotification</code> to avoid N+1 issues.
NotificationResponseDTO	Unified schema for frontend consumption, including <code>linkedId</code> and <code>type</code> .

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